

Case Study - PatchPad Editor v3

A long-report editor combining exact text operations with durable, conflict-safe revisions

Harbor task: turing/patchpad-editor-v3 | version 1.0.0 | prepared 13 September 2026

Scope: recorded platform outcomes. GPT-5.4 mini is the score target; Gemini has no score restriction. The user authorized final packaging with the reported evidence limitations retained. No scores, task requirements or verifier weights were changed.

Recorded scores: Oracle 1.0000 | GPT-5.4 mini 0.5605 | Haiku 4.5 0.2233 | Gemini 3.7 Flash 0.7622 | no-op 0.0000

1. What the task is

Build PatchPad from an empty /app: a browser editor for long incident reports, with a custom editing surface, SQLite as the source of truth and a reliable saved-revision history. The supplied Northwind API Incident Report has 1,226 logical lines, including exact generated markers and a tail sentinel. Five detailed instruction files accompany the short product brief.

Area	Required behavior
Custom editor and seed	Custom DOM/canvas/SVG document surface; supplied Northwind API Incident Report; 1,226 logical lines, original metadata and revision 1.
Editing and selection	Logical line/cursor positions, word navigation, grapheme-safe Unicode, mouse word/line/range selection, offscreen selection and multiple carets.
Clipboard and history	Exact multiline/tabbed copy-cut-paste, typing groups, atomic selection replacement, block indentation, Replace All, Undo/Redo and redo-branch invalidation.
Saved revisions	Explicit Save, no-op save invariants, reload/fresh-client persistence, revision preview and unsaved restore with Undo.
Server safety	SQLite durability, restart seed idempotence, chained two-tab stale-save conflicts and malformed/contradictory save rejection without mutation.
Interface	Discoverable controls, focus and readable feedback; five visual axes at 1280 by 800. No mobile or sign-in requirement.

One Node.js application starts with `npm start` on `0.0.0.0:${PORT:-3000}`. Dependencies must be available at startup. Public networking is permitted; document data, saves and history remain local and server-backed. A native `textarea/contenteditable` or ready-made editor may not implement the document surface; a small hidden input for keyboard or clipboard plumbing is permitted.

2. Why the task requires more than presentation

The same document must survive distinct state transitions: exact unsaved edits, grouped Undo/Redo, persisted saves, revision restoration and concurrent writes. A visually convincing editor can still split graphemes, change an unselected line during indentation, lose an Undo group or overwrite a newer revision.

Version v3 adds selected-block indentation with preserved selection and atomic history, plus whole-document Replace All Undo/Redo and branch invalidation. These behaviors are stated in the product brief and each has a distinct Functional criterion. The existing 60/20/20 reward formula remains unchanged.

3. How it is evaluated

Codex with gpt-5.6-luna and max reasoning uses Playwright MCP against the live app. There are 41 criteria: Render 2, Constraints 2, Functional 29, Polish 3 and Visual 5. Functional criterion weights total 22.75. Render/Constraints are

prerequisites; the final reward is 60% Functional, 20% Polish and 20% Visual. The agent budget is 7,200 seconds and the separate verifier budget is 13,200 seconds.

Run	Overall	Render	Constr.	Func.	Polish	Visual	Recorded status
Oracle	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	29/29 Functional
GPT-5.4 mini (high)	0.5605	1.0000	1.0000	0.4286	0.6667	0.8500	12/29 Functional
Gemini 3.7 Flash (high)	0.7622	1.0000	1.0000	0.6703	1.0000	0.8000	22/29 Functional
Claude Haiku 4.5	0.2233	1.0000	1.0000	0.0000	0.6667	0.4500	Functional gate failed; 29 checks skipped
No-op control	0.0000	0	0	0	0	0	graded=0; no_op=1

4. Target-model result: GPT-5.4 mini

GPT-5.4 mini with high reasoning completed normally, passed Render and Constraints, and scored 0.5605 overall. Functional was 0.4286 (12/29 criteria; 9.75 of 22.75 weight), Polish 0.6667 and Visual 0.8500. Its official reward is within the 0.10-0.70 target band. The internal 0.50 aspiration is not an additional acceptance restriction.

The judge reported unpadding seed markers, incorrect selection/navigation outcomes and block indentation that joined BLOCK-THREE with BLOCK-OUTSIDE. It also credited persistence, chained stale-save conflicts, revision preview/restore, restart durability and the direct API rejection matrix. The run therefore demonstrates substantial implementation alongside reported editing defects.

Five editing/persistence criteria also contain incomplete setup or evidence capture. These are not all proven judge mistakes: an app defect may have prevented setup. They were not reproduced against GPT during this preparation. Missing padded seed markers also block dependent tests; failed criteria should not be counted as separate root defects. The recorded score is preserved, not replaced with an estimated corrected score.

Recorded criterion	Evidence limitation
edit save reload and fresh client	The marker saved and reloaded, but the full pre-save draft comparison was not established.
backspace delete exact line join	Two attempts did not establish the required three adjacent lines, so the boundary edits were unverified.
multi caret full typing single undo	The judge did not establish three visible carets, so typing/Undo/Redo was unverified.
multi caret backspace delete sibling	The judge did not establish the second caret required for deletion.
replace all atomic history and branch	Replace All and single Undo/Redo worked, but the branch insertion and Redo invalidation checkpoint were not established.

5. Oracle and comparison runs

Oracle scored 1.0000 in every dimension and passed all 41 criteria, including all 29 Functional criteria. The empty-submission control scored 0.0000 with graded=0 and no_op=1. The Oracle source exported by the run matches the packaged golden application.

Gemini completed with no trial exception and scored 0.7622. It has no score restriction and is included as comparison evidence. Its judge lost a Restore/Undo checkpoint to a result-assembly error. A focused local browser diagnostic on unchanged source passed revision preview, unsaved Restore and one Undo, while independently reproducing multi-caret typing Undo leaving MULT after typing MULTI. These diagnostics do not change the official score or establish a full regrade.

Haiku completed at 0.2233, with Render/Constraints 1.0, Functional 0.0, Polish 0.6667 and Visual 0.45. Functional reported a zero-width/clipped editor and failed its browser gate, skipping all 29 behavior checks. Other dimensions credited Find/preview interactions. Its total is comparison evidence, not 29 independently demonstrated behavioral failures.

6. What these results support

The official Oracle-to-GPT gap is 0.4395. Oracle completed all Functional requirements; GPT received stronger presentation scores than Functional scores. This supports reporting a useful measured separation for the target model, while the identified evidence gaps limit causal claims about individual failures. One trial per model is not a repeatability study or complete fairness certification.

Delivery and provenance

The final folder follows the supplied seven-file delivery layout: unchanged task ZIP, Oracle/no-op job-directory ZIP, GPT-mini job-directory ZIP, Gemini job-directory ZIP, Haiku job-directory ZIP, this case study and the companion evaluation report. Job exports preserve recorded scores, criterion reasons, source and trajectories. Runtime databases and caches are excluded; sanitization is packaging-only.

All five trials share task checksum 132a2f0fb7ae8e5a9a02f154fe7d5de1d2c232c7aaf45e3e3f0c5d057dc9f659. All logged verifier hashes match the packaged source. All 32 task archive files match current source; the task ZIP checksum is a3511a525675554aaa1ee89b5e77fb69ad4f7a94a1fc68b7d137bc39053439a6.

No deliberate cheating was found in the reviewed source and builder trajectories. The scan found no matching credential tokens or grader-directed tampering. This is a scoped assessment, not an absolute guarantee. Native judge action traces and a platform QC report were not present in the supplied exports. Packaging does not claim either has been produced.

The user authorized this final package after review of GPT evidence gaps. Scores and requirements remain unchanged; Gemini is not subject to the target score band. The evaluation report contains every criterion outcome and the recorded failure explanations, with an index to the included jobs.